



Volume and thickness change of NMC811|SiO_x-graphite large-format lithium-ion cells: from pouch cell to active material level

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HIGHLIGHTS

- Reversible thickness change of large format silicon-containing cell measured.
- Electrode dilatometry of silicon-graphite composite and nickel-rich cathodes.
- Nonlinear expansion of nickel-rich cathodes impacts large-format cell expansion.
- Validation of expansion model of silicon and graphite using dilatometry.
- Towards a targeted battery cell and module housing design for electric vehicles.

ARTICLE INFO

Keywords:

Lithium-ion batteries
SiO_x-Gr composite anode
Ni-rich cathode
Pouch cell expansion
Silicon expansion
Dilatometry

ABSTRACT

In this study, the reversible thickness change of a large-format lithium-ion automotive pouch cell is investigated under precisely monitored cell pressure and temperature using an in-house built actively controlled pneumatic cell press. The quantitative and qualitative contribution of the state-of-the-art NMC811 cathode and the SiO_x-graphite composite anode to the total pouch cell expansion is resolved by electrochemical dilatometry and validated. Results show that Ni-rich cathodes have a significant impact on the pouch cell expansion and exhibit highly nonlinear thickness change which is related to the change of the individual lattice parameters of the crystal structure. To resolve the contribution of both anode active materials to the total anode expansion, the capacities of SiO_x and graphite are determined by differential voltage analysis and validated with half-cell measurements. Then, the volume expansion of SiO_x and graphite as a function of the anode state of charge is calculated. By introducing fitting parameters and applying theories about the interaction of SiO_x with the surrounding morphology the correlation between the volume expansion of the active materials and the thickness change of the SiO_x-Gr composite anode is investigated. The findings suggest that there is significant nonlinear reduction of pore volume at low state of charge.

1. Introduction

Silicon as an active material is becoming increasingly common in the anode of modern lithium-ion (Li-ion) cells [1,2]. Particularly, in the automotive industry active materials with high capacities are of great interest, since volumetric energy density is a key design parameter due to limited available space in electric vehicles (EV). Silicon with its high theoretical capacity of 3579 mAh g⁻¹ and material abundance offers opportunities but also challenges [3–5]. One of the most critical

challenges is the large volume expansion upon lithiation [6–8]. Even if silicon nano-particles are used, the expansion may lead to particle cracking if the particle size is above 150 nm [9]. Particle cracking leads to electrical disconnection and additional growth of the solid electrolyte interphase (SEI) and thus loss of cyclable lithium [5,9]. Willenberg et al. showed that these processes lead to jelly roll deformation in cylindrical cells [10–12]. Among other challenges, such as inferior electrical conductivity [5], silicon anode material research for

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the application in Li-ion cells thus put great effort into the mitigation of the effects of expansion [9,13–17]. One such approach is to use silicon oxide (SiO_x) rather than pure silicon. During the first lithiation of SiO_x a Li_4SiO_4 -matrix with embedded Si-particles is formed. While the Li_4SiO_4 -matrix is inert and inactive solely the embedded Si-particles are electrochemically active [18,19]. However, the Li_4SiO_4 -matrix acts as a buffer for the expanding Si-particles so the volumetric expansion of SiO_x is much lower than that of pure Si (118% vs. 280%) [19]. In combination with other advantages like higher electrical conductivity and faster Li-diffusion SiO_x thus is a promising candidate to combine the advantages of high volumetric capacity yet lower volume expansion compared to pure Si [3,19]. For the cathode $\text{Li}(\text{Ni}_x\text{Co}_y\text{Mn}_z)\text{O}_2$ (NMC) is a commonly used high-energy material due to its high voltage level [20, 21]. In recent years the Ni-content has been increased continuously in order to reduce the share of the expensive materials Mn and Co and make use of the high energy density of Ni [22]. However, increasing the Ni-content also results in greater structural instability [23], a significant collapse of the lattice parameter c of the rhombohedral crystal structure during delithiation and larger volume change [20,24]. Above mentioned challenges for increasing the silicon content thus also apply for increasing Ni-content in NMC [20,24].

Inside the battery system pouch cells are braced in the industrial assembly process which ensures mechanical integrity and stability with regard to vibration and crash behavior [25]. Bracing leads to mechanical tension in stacking direction and the cells experience a compression force. This is important for laminated pouch cells as unconstrained cycling leads to layer delamination [26,27]. Applying low pressure further improves electrical contact [28] and expels gas from the electrode stack [29,30]. On the other hand, high stack level stress leads to higher rates of capacity fade due to localized separator deformation and chemical degradation [26,31]. Ideally, the stress level would be set to a certain constant value of choice during the module assembly process that ensures mechanical integrity and minimizes stress level dependent degradation. However, when the cell thickness changes reversibly due to electrode lithiation and delithiation the pressure in the module and the compression force on the cells deviates from the initial value according to the stiffness of the module housing [25,32]. Forces induced by thickness growth may also lead to lengthening and damaging of the module housing [32]. This is especially significant if active materials that show large volume changes during cycling like Ni-rich NMC cathodes and Si-containing anodes are used as explained above. To design optimal electrodes, cells and modules the reversible volume and thickness changes must be fully understood — from pouch cell to active material level. Note that cell thickness may also grow irreversibly due to aging such as SEI growth [33,34]. This has been investigated in previous studies [25,32,35] and this work is explicitly designed to rule out all irreversible processes and focus on reversible processes only.

Various studies have investigated the reversible thickness change of Li-ion cells during cycling with Si-graphite (Si-Gr) composite and SiO_x -graphite (SiO_x -Gr) composite anodes. Louli et al. have shown that the intrinsic voltage hysteresis of Si causes large volume hysteresis between lithiation and delithiation [36]. However, the correlation between volume expansion and thickness change of the anode was not further investigated. Vidal et al. [37] have calculated the porosity change of SiO_x -Gr composite anodes by subtracting the theoretical thickness change from the measured thickness change but they did not differentiate between the contribution of SiO_x and graphite to the thickness and porosity change. For Ni-rich cathodes, large volume expansions of >5% inside the crystal structure of NMC811 have been reported [20,23,24]. Recent findings on the other hand observed only very minor thickness change of <0.5% [38] which has not yet been confirmed on pouch cell level. Previous studies have also mostly used small lab-scale pouch cells to investigate thickness changes [28,36,39]. This leaves a degree of uncertainty when transferring results onto much

larger pouch cells with a magnitude of difference in size and capacity as they are common in automotive applications.

To the best of our knowledge, to this date, no single study exists that precisely investigates and validates the volume expansions and thickness changes on all macroscopic and microscopic levels. In this work this gap is closed to generate a clear understanding of how volume changes of active material particles propagate through the thickness changes of individual electrodes to the overall thickness change of the pouch cell under realistic operating and bracing conditions. To achieve this the following approach is taken which is visualized in Fig. 1:

- Large-format NMC811| SiO_x -Gr pouch cells with SiO_x are cycled under precisely controlled operating conditions until no more irreversible thickness change is detectable within the sensor resolution. The temperature is held constant inside a climate chamber. The pressure on the cell is held constant regardless of displacement by an actively-controlled pneumatic cell press and the cell thickness is measured.
- Dilatometer measurements of individual anodes and cathodes are performed. The results of the dilatometer measurements are validated with the results of the cell-press measurements.
- The validated cathode thickness change is analyzed regarding the change of the lattice parameters and the unit cell volume of the rhombohedral crystal structure of NMC811 as well as previous dilatometer measurements [38].
- Half-cell voltage curves of the anode are measured. The corresponding voltage curves of SiO_x and graphite are calculated based on own measurements with pure SiO_x and pure graphite electrodes and differential-voltage-analysis. The calculated theoretical voltage curve of the anode is validated with the actual measurement and the SOC as well as the volume expansion of SiO_x and graphite as a function of the composite anode SOC is calculated.
- The correlation between the volume expansion of SiO_x and graphite and the validated thickness change of the composite anode is investigated starting with the theory of constant pore volume [31,40] and subsequently introducing additional fitting parameters and theories about the microscopic interactions of the active materials with the surrounding materials. The resulting theoretical contributions of SiO_x and graphite to the anode thickness change is validated with the dilatometer measurements.

2. Experimental

2.1. Investigated cell

A large-format automotive pouch cell with >60 Ah, composite SiO_x -Gr anodes with SiO_x content between 8–12 wt.-% and Ni-rich NMC811 cathodes was investigated. Current collectors and gas pockets were located on opposite sides. One-sided anode coating thickness was in the range of 65–75 μm , one-sided cathode coating thickness in the range of 55–65 μm and the number of individual layers in the range of 32–36. The exact values are known to the authors but are subject to confidentiality and not required for the investigation.

2.2. Electro-mechanical testing

An actively controlled pneumatic press (Fig. 1 top-left, from here on referred to as ‘cell press’) was developed in-house to enable thickness measurements of large-format pouch cells for automotive applications under precisely controlled operating and pressure conditions. The cell press used in this work and its various advantages over conventional thickness measurement equipment are extensively presented and described in detail by Deich et al. [25] so only a brief description will be given here. Four metallic plates are arranged parallel on a vertical axis. Plates 1 and 4 are rigid while plates 2 and 3 can move in

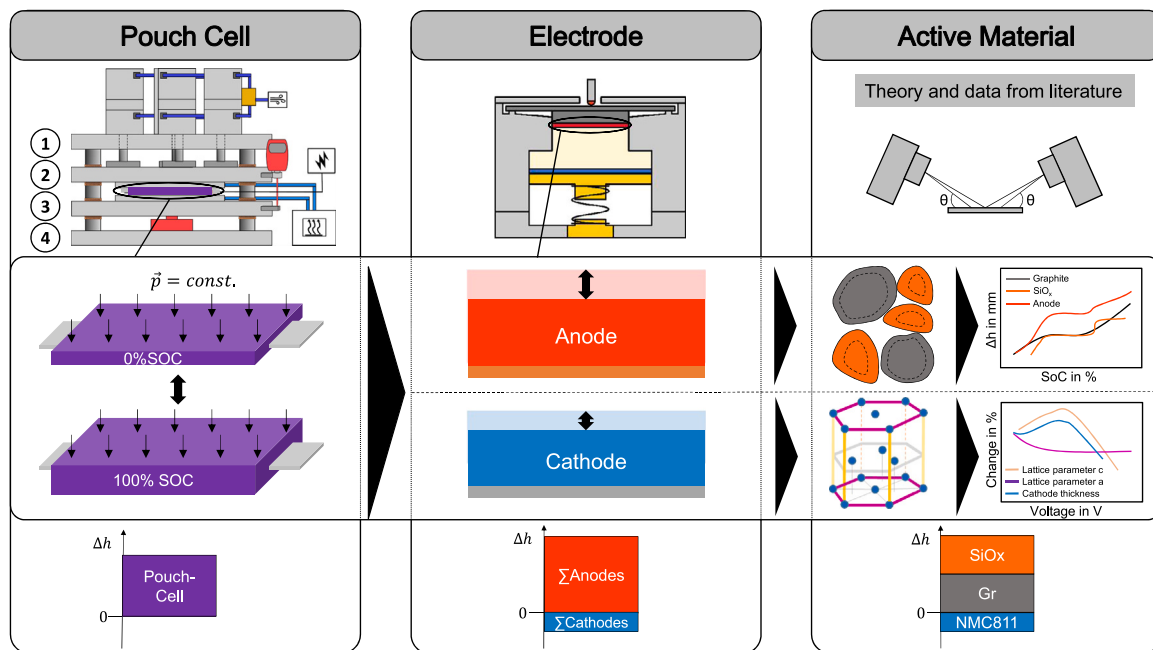


Fig. 1. Overview of the approach taken in this work to investigate the thickness and volume changes on all macroscopic and microscopic levels. Measurement of the thickness change on pouch cell level (left), measurement of the thickness change on electrode level (middle) and theoretical investigation of the relationship between volume change of the active materials and the thickness change of the respective electrode (right). Pouch cell measurements are used to validate the electrode measurements, electrode measurements are used to fit and validate the relationship between volume change of the active materials and the thickness change of the electrodes.

vertical direction. Cells are fixed between two cooling plates that are mounted on the top and the bottom of plates 2 and 3, respectively. Four pneumatic cylinders exert a user-defined force through plate 3 onto the cell either before, during and/or after operation (constant or as a function of time or displacement).

Before cycling the cell pressure was set to 0.5 MPa which is a common value inside vehicles battery housings [32]. The pressure is sufficient to squeeze gases out [29] and to ensure flattening of the electrode layers perpendicular to the pressure plate. At the same time, it is low enough to have no detrimental effects, such as accelerated capacity fade [32,41]. After initial compression the cell was then allowed to rest for 70 h before cycling to ensure mechanical retardation processes of the inactive components like the aluminum pouch-bag and the separators become insignificant afterwards. During cycling the cell pressure was held constant at 0.5 MPa regardless of displacement. All measurements were performed inside a climate chamber (VC3 4100, Vötsch Industrietechnik GmbH) which was set to a constant temperature of 25 °C. No additional liquid surface cooling through the cooling plates was applied because the small operating currents only generated minor heat which was quickly absorbed by the large heat capacity of the metal plates. A HPS10 from BaSyTec GmbH was used for electrochemical measurements. Some irreversible thickness change was observed within the first few cycles in previous experiments with cells of the same kind. To generate a representative cycle for reversible thickness change only, first, 20 cycles with a current of C/3 were performed to ensure irreversible thickness change becomes neglectable. Then, the current was set to C/10 at the beginning of discharging of cycle 21 and afterwards three more cycles were carried out to precisely capture all details of the thickness change and to show repeatability. For all cycles a constant current (CC) discharge protocol to 2.75 V and constant current constant voltage (CCCV) charge protocol to 4.20 V with a cut-off current of C/20 was applied.

2.3. Dilatometry

The thickness change of single one-sided anode and cathode coatings was measured by EL-Cell GmbH using an ECD-3-nano type

dilatometer (Fig. 1 top-middle). The dilatometer is designed as a half-cell setup where the working electrode is cycled against lithium. The sensor unit can obtain displacement signals of <5 nm. The working electrode (10 mm in diameter) and the counter electrode (lithium metal, 12 mm × 0.3 mm, Sigma Aldrich) are separated by a porous glass separator to guarantee that only thickness change of the working electrode is measured. All parts were dried overnight at 80 °C in vacuum and the glass separator was additionally dried overnight at 170 °C in vacuum. The dilatometer cell was assembled in an Ar-filled glovebox (M. Braun Inertgas-Systeme GmbH, H₂O < 1 ppm, O₂ < 1 ppm). 345 μL 1M LiPF₆ in EC/DEC (50:50 v/v) (Sigma-Aldrich) with 5 wt.-% FEC (99 %, Sigma-Aldrich) was used as electrolyte. All tests were performed at 25 °C inside a climate chamber (Binder GmbH) with a Biologic VSP multichannel impedance analyzer. Anodes were cycled for 3 cycles without resting between charging and discharging between 0.05 V and 1.00 V vs. Li/Li⁺ with a CCCV protocol with C/10 and C/100 cut-off current. Cathodes were cycled for 5 cycles between 3.00 V and 4.20 V vs. Li/Li⁺ with a CCCV protocol with C/20 and C/100 cut-off current.

2.4. Laboratory test cells

Electrodes for the use in laboratory test cells were gained from a fresh cell after formation. The cell was discharged to 2.00 V and opened in an Ar-filled glovebox (GS Glovebox Syntemtechnik GmbH, H₂O < 0.1 ppm, O₂ < 0.1 ppm). Anodes and cathodes were rinsed in dimethyl carbonate (DMC, Sigma-Aldrich) twice for 12 h each. The coating was carefully removed on one side using 1-Methyl-2-pyrrolidone (NMP, Sigma-Aldrich). The test of half- and full cells of small laboratory formats were carried out with a CTS Lab XL cell tester (Basytec GmbH) in case of the ECC-Std. tests cells (EL-Cell GmbH) and in a climate chamber (Weiss Umwelttechnik GmbH) at 25 °C. For PAT-Cell (EL-Cell GmbH) measurements a PAT-Standard-16 (EL-Cell GmbH) in a climate chamber (Binder GmbH) at 25 °C was used. Both cell types use electrodes with a diameter of 18 mm, which were punched out using a special tool (EL-Cut by EL-Cell GmbH). The voltage curves and areal capacities of the anodes and cathodes were

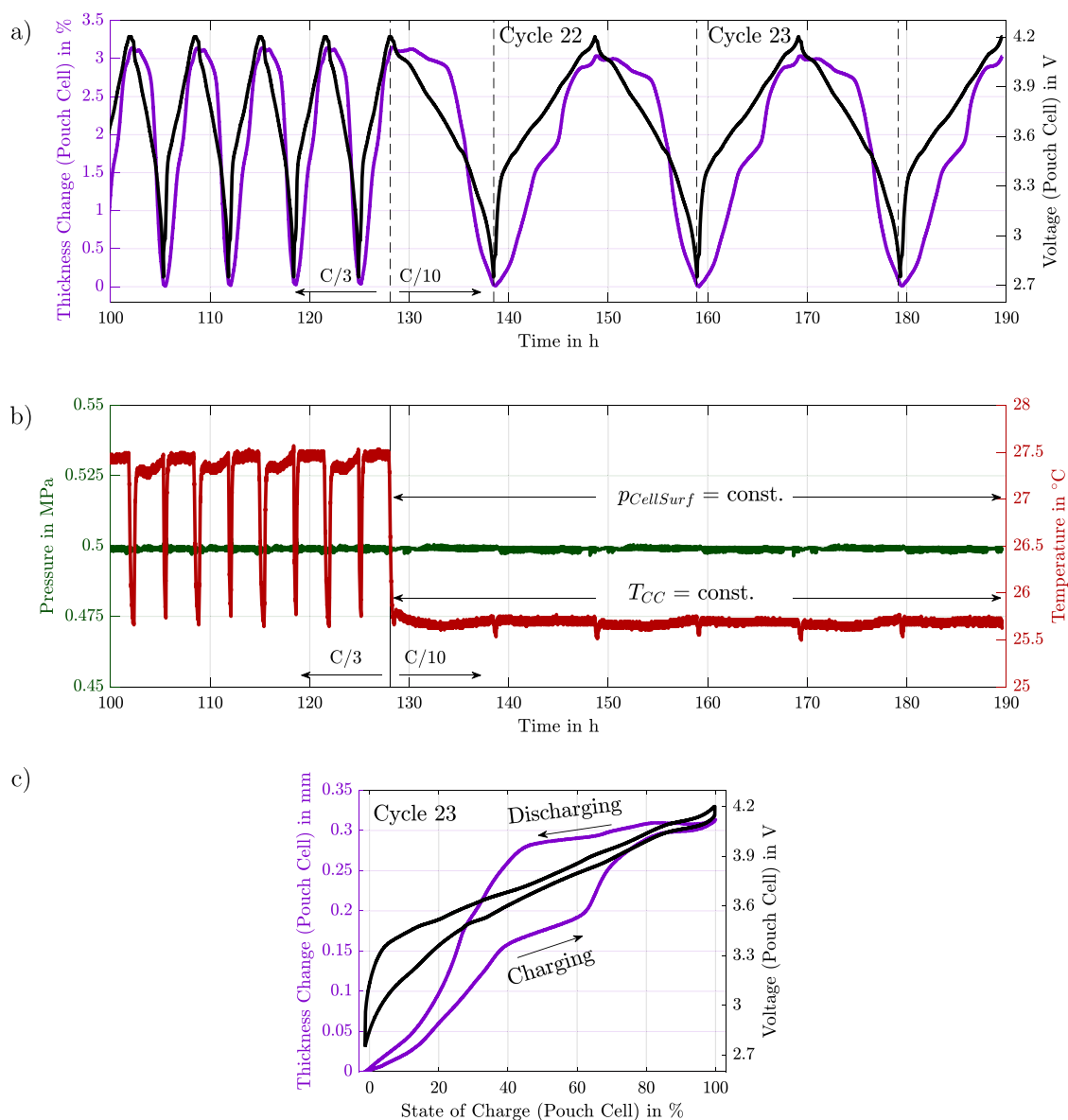


Fig. 2. Electro-mechanical measurements with the actively-regulated pneumatic cell press. (a) Percentage pouch cell thickness and voltage change and (b) cell pressure and temperature on current collector over time for cycles 17 to 24 with C/3 until beginning of discharging of cycle 21, afterwards C/10. (c) Pouch cell thickness change and voltage over SOC for cycle 23.

gained by cycling the respective electrode against Li/Li^+ with a low current of C/120. The anode was cycled between 1.00 V–0.05 V and the cathode between 3.00 V–4.30 V. Balancing was gained by full cell measurements with Li-reference between 2.50 V–4.20 V. Graphite and SiO_x half-cells were produced in-house. The anode slurries were made using a Thinky ARE 250 speed-mixer (Thinky Corporation). The graphite anode consisted of 94 wt.-% graphite and each 2 wt.-% styrene-butadiene rubber (SBR), carboxymethyl cellulose (CMC) and carbon black (CB). The SiO_x anode consisted of 82 wt.-% SiO_x with each 7 wt.-% SBR and CMC and 4 wt.-% CB. Using a doctor blade (ZUA 2000.100, Zehntner GmbH) and an automated film applicator (Coatmaster 510, Erichsen GmbH & Co. KG) single-sided loading of 3.50 mAh cm^{-2} was set for both graphite and SiO_x anodes. Porosity was set to 55% for the SiO_x anode and 38% for the graphite anode using a SUMET CA9/250-200 calander (Sumet Systems GmbH).

3. Results and discussion

3.1. Pouch cell expansion

Fig. 2(a) shows pouch cell thickness and voltage over time of cycles 22–24 that were carried out with C/10 to investigate the reversible thickness change and four of the 20 previous cycles with C/3 that were performed to ensure all measureable irreversible thickness change happened before cycles 22–24. The cell thickness for the 3 cycles with C/10 at 0% SOC and 100% SOC were almost identical in a range of 0.001 mm. Fig. 2(b) shows cell pressure and temperature on the surface of the current collectors for the same time span. The measured temperature on the surface of the current collectors for cycles with C/3 ranged between 27.5 °C and 25.6 °C and for cycles with C/10 between 25.7 °C and 25.5 °C. The measured pressure on the cell for all cycles ranged between 0.501 MPa and 0.497 MPa. This indicates that the cell

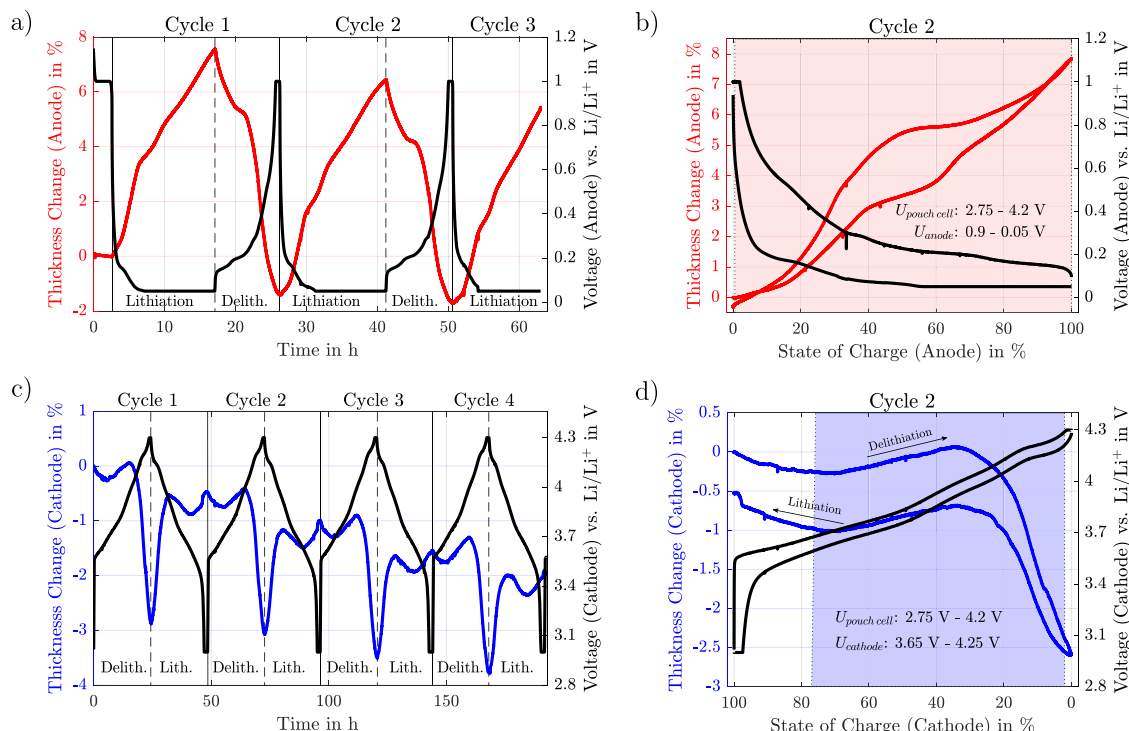


Fig. 3. Dilatometry measurements with an ECD-3-nano dilatometer by EL-Cell GmbH. (a) Anode thickness and voltage for all cycles over time, (b) Anode thickness and voltage of cycle 2 over SOC, (c) Cathode thickness and voltage for all cycles over time, (d) Cathode thickness and voltage of cycle 2 over SOC. Anode measurements with C/10, cathode measurements with C/20, both with a CCCV protocol with C/100 cut-off-current. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

press was able to control the operating conditions in such a way that only reversible thickness change caused by lithiation and delithiation was measured and all further influence was successfully excluded. That said, the C/10 cycles can be considered suitable for the investigation of thickness change caused by reversible lithiation and delithiation processes only under constant pressure and constant temperature. The 23rd cycle was chosen for further investigation. Fig. 2(c) shows pouch cell thickness change and voltage for cycle 23 over SOC with respect to the capacity of this cycle. During charging the cell thickness increased, during discharging the cell thickness decreased. At 100% SOC the cell thickness was 3.03% greater compared to the cell thickness at 0% SOC. There was pronounced hysteresis in voltage and thickness between charging and discharging. At the beginning of discharging the cell thickness first decreased and then increased again. In the following sections the cause of these observations will be meticulously tracked down to the electrode and further down to the active material level.

3.2. Electrode to pouch cell expansion

Fig. 3(a) shows anode thickness and voltage over time measured in the dilatometer. For demonstration of repeatability two full cycles and the beginning of lithiation of the third cycle are shown. The curve shows good repeatability with a slight decrease of anode thickness at the end of delithiation most likely caused by the drift of the sensor used in the dilatometer. Overpotentials and long CV-phases caused the lithiation phase to take longer than the delithiation phase. For further in-depth investigation the second anode cycle was chosen (Fig. 3(b)). Lithiation capacity was 2.66 mAh, delithiation capacity was 2.61 mAh which calculates to an efficiency of 98.1%. When cycled between 1.00 V and 0.05 V the anode experienced a thickness change of 7.85%. When the pouch cell is cycled between 2.75 V and 4.20 V the anode is cycled between about 0.90 V and 0.05 V (Fig. S1). Thus, as marked in Fig. 3(b) in light red, only this voltage range is used for further investigation. The decreased voltage range reduced the capacity by 1.5% to 2.62 mAh (charging) and 2.57 mAh (discharging).

Fig. 3(c) shows cathode thickness and voltage over time. For demonstration of repeatability four cycles are shown. The thickness change shows good qualitative repeatability both in lithiation and delithiation. In absolute numbers each delithiation phase looks similar to the previous lithiation phase, finishing delithiation slightly lower than at the beginning of the previous lithiation. On the other hand, each lithiation phase fails to catch up to the thickness level at the beginning of the previous delithiation phase. All cycles had an efficiency between 97.66% and 97.99%. According to the manufacturer of the ECD-3-nano dilatometer (EL-Cell GmbH) who executed these experiments, it is more difficult to measure cathode thickness change than anode thickness change because the cathode is more vulnerable to water residue inside the dilatometer. It is likely that the continuous decline in cathode thickness does not indicate measurement errors but is rather caused by the reasons stated above. The validity of this assumption will be proven in a later section. For further investigation the second cathode cycle is chosen. Charging capacity was 3.28 mAh, discharging capacity was 3.21 mAh which calculates to an efficiency of 97.9%.

Fig. 3(d) shows cathode thickness over SOC for the second cycle. Unlike the anode, the cathode thickness does not rise or fall monotonously but is highly nonlinear. This behavior has been linked to the collapse of the lattice parameter c of the rhombohedral crystal structure [20,42]. In agreement with [20] a very strong thickness change was observed when the voltage exceeded 4.00 V. When cycled between 3.00 V and 4.30 V the cathode experienced a thickness change of 2.58%. Detailed analysis of the change in the crystal structure, the unit cell volume and the corresponding thickness change is conducted in Section 3.3.1.

When the pouch cell is cycled between 2.75 V and 4.20 V the cathode is cycled between 3.65 V and 4.25 V (Fig. S1). Again, as marked in Fig. 3(d) in light blue, only this voltage range is used for further investigation. Between 3.65 V and 4.25 V the thickness changed by 2.11%. The resulting capacity in this range was 2.46 mAh (charging) and 2.41 mAh (discharging).

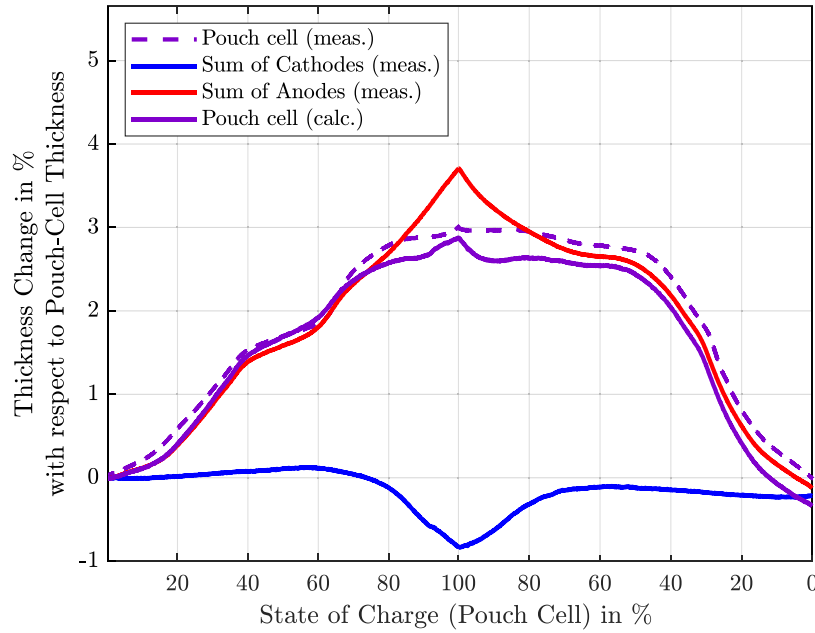


Fig. 4. Sum of the thickness changes as a percentage of the pouch cell thickness at 0% SOC for all anode coatings (solid red) and all cathode coatings (solid blue) calculated from dilatometer measurements of a single one-sided coating of anode and cathode, respectively. The sum of both anode and cathode thickness changes equals the calculated thickness change of the pouch cell (solid purple). The comparison with the measured thickness change of the pouch cell measured in the cell press (dotted purple) shows good quantitative and qualitative agreement. Unsymmetric thickness change of the cathode between delithiation and lithiation (Fig. 3(c)) leads to an offset in absolute values for discharging. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

To validate the dilatometer measurements, the assumption is made that the thickness change of the pouch cell is the sum of the total thickness changes of all individual electrodes (Eq. (1)). Since the pressure on the cell was held constant by the cell press regardless of displacement (Fig. 2(b)) and enough time was given for mechanical retardation processes of the inactive components prior to the measurements we believe this is a valid assumption. The total electrode thickness change is therefore the thickness change of a single electrode with two coatings multiplied with the number of layers (Eqs. (2) and (3)).

$$\Delta h_{\text{Pouchcell}} = \sum \Delta h_{\text{Electrodes}} = \Delta h_{\text{Anode,total}} + \Delta h_{\text{Cathode,total}} \quad (1)$$

$$\Delta h_{\text{Anode,total}} = 2 \cdot \Delta h_{\text{Anode,coating}} \cdot n_{\text{Anode,layers}} \quad (2)$$

$$\Delta h_{\text{Cathode,total}} = 2 \cdot \Delta h_{\text{Cathode,coating}} \cdot n_{\text{Cathode,layers}} \quad (3)$$

The investigated pouch cell had one more layer of anodes than layer of cathodes. However, the very outer anode coatings on both sides were not included in Eq. (2) since they were not lithiated during cycling and thus do not experience thickness change. Fig. 4 shows the total thickness change of all coatings for both electrodes calculated according to Eq. (2) and Eq. (3) as a percentage of the pouch cell thickness at 0% SOC, the theoretical thickness change of the pouch cell calculated with Eq. (1) and the measured pouch cell thickness change in the cell press.

The anode curve follows the characteristics of the measured full cell curve closely until about 65% SOC. Especially the beginning and end of the graphite-plateau between 40% SOC and 60% SOC make a good indicator that the anode curve is placed correctly on the x-axis and the voltage range of 0.90 V–0.05 V vs. Li/Li⁺ for the anode was identified correctly. The measured pouch cell curve deviates from the anode curve and flattens at about 65%–70% SOC. This is caused by a steep decline of the cathode curve beginning in this region which again indicates that the voltage range of the cathode of 3.65 V–4.25 V vs. Li/Li⁺ was measured correctly. At about 90%–95% SOC the measured pouch cell curve starts to rise again which is caused by the flattening of the cathode curve. For discharging similar to charging there is

again deviation between pouch cell and anode curve until about 70%–65% SOC because of strong thickness change of the cathode. From this point onwards the change in cathode thickness becomes small and the pouch cell curve becomes closely related to the anode curve again. The total thickness change of all anode coatings calculates to 3.70% of the pouch cell thickness at 0% SOC, the total thickness change of all cathode coatings calculates to –0.83% of the pouch cell thickness at 0% SOC. The sum of the anode and cathode thickness changes calculate to the theoretical thickness change of the pouch cell of 2.87%. The total thickness change measured in the cell press was 3.03%. For discharging the calculated values are smaller than the measured values with a rather constant offset which was to be expected because of the smaller thickness change of the cathode for lithiation as discussed in Section 3.2. Overall, considering the totally different measurement equipment used for pouch cell and electrode measurements and the difficulty of accurately measuring the thickness change of the cathode, the calculated values are in good agreement with the measured values. Thus, the dilatometer measurements of the anode and cathode can be viewed as validated and representative for the behavior of the respective individual electrode inside the pouch cell and are suitable for further investigation.

To sum up, the NMC811 cathode counteracted the thickness change of the SiO_x-Gr composite anode by 21.8%. For automotive battery design this is very pleasant behavior as less reversible thickness change means more available space for energy-storing material. That said, Ni-rich cathodes not only increase the total battery energy content for a given design space directly by providing higher specific energy per NMC-volume but also indirectly by reducing the thickness change on pouch cell level. Less reversible thickness change further makes it easier to operate the cell at optimal pressure [26] for maximum cycle life. By shifting the electrode balancing and increasing the upper cathode cut-off voltage beyond 4.25 V it is theoretically possible to counteract the thickness growth of the anode even further as the cathode thickness change gradient remains large beyond this voltage level (Fig. 3(c)) in agreement with [42].

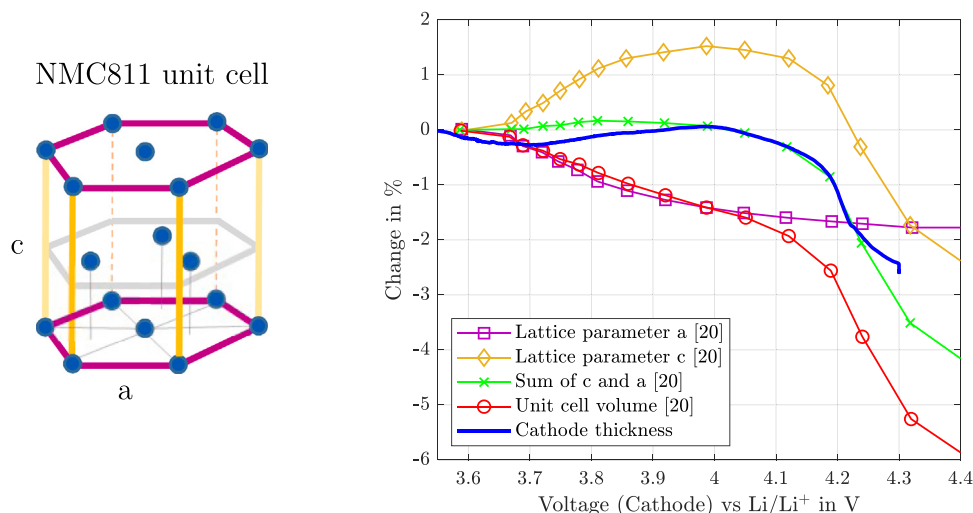


Fig. 5. NMC811 unit cell (left) and thickness change of the cathode compared to the change of the lattice parameters a and c as reported in the literature [20], the sum of a and c and the unit cell volume (right). A similar investigation was performed recently with different results [38].

3.3. Active material particle to electrode expansion

3.3.1. Cathode

NMC811 crystallizes in a rhombohedral $R3m$ -structure [20]. Fig. 5 shows the NMC811 unit cell and the change of the lattice parameters a and c as reported by Biasi et al. [20], the change of the sum of a and c , the change in unit cell volume calculated as $V = a^2 \cdot \sin(60^\circ) \cdot c$ and the cathode thickness change for delithiation measured in this paper. The measured thickness change of the cathode does not follow the change of the unit cell volume but is rather related to the change of the lattice parameter c . The peak in cathode thickness is measured at 3.95 V which also exactly matches the peak of the lattice parameter c . Afterwards the cathode thickness starts to decrease rapidly similar to the decrease of the lattice parameter c . Interestingly, the measured cathode thickness change follows the sum of a and c almost perfectly between 4.00 V and 4.25 V. When delithiated to 4.30 V the cathode thickness decreased by 2.11% while the unit volume decreased by about 5.00%. This indicates that a significant portion of the volume contraction leads to porosity change rather than thickness change. The validated thickness change of 2.11% between 3.65 V and 4.25 V vs. Li/Li⁺ measured in this study stands in contrast to the thickness change of nearly 0% within the same voltage range for a different NMC811 cathode measured in a previous study [38]. This is an important finding as the cathode counteracts the increase in thickness of the anode (Fig. 4) and decreases the overall thickness change on pouch cell level. A larger thickness change of the cathode thus allows more active material to be stored inside a given space as discussed in Section 3.2. From this perspective engineers should be aware of the fact that different NMC811 cathodes may show completely different expansion behavior and take this aspect into account when designing electrodes and cells.

3.3.2. Anode

In the composite anode two different active materials are present, SiO_x and graphite. In this section, the validated anode expansion is further investigated to identify the contribution of both active materials to the expansion of the composite anode. To do so, knowledge of the capacity contributions of both active materials is required. In theory, the capacities can be calculated by multiplying the known total active material masses with the theoretical gravimetric capacities. However, the exact stoichiometry and the initial irreversible capacity loss of SiO_x are unknown and both active materials are never fully lithiated or delithiated. Thus, in this paper we chose to identify the active material capacities by performing DVA of the anode discharge voltage curve vs. Li/Li⁺ which leads to accurate results [43,44]. The validity of such

procedure has been proven by Berhaut et al. [45] by x-ray synchrotron measurements and will also be validated later in this section.

Capacity estimation

Fig. 6(a) and (b) show half-cell voltage curves and DVA of discharge over capacity for the anode. In Fig. 6(b) distinct peaks can be related to different active materials [44]. During discharge, most of the capacity of graphite is released before silicon starts delithiating [36]. Using this knowledge it is possible to estimate the capacity contribution of silicon and graphite to the total capacity of the composite anode using DVA [43]. The distinct silicon peak arises at 3.3 mAh with a separation voltage of 0.29 V indicating silicon delithiation happens above this voltage level. With the assumption that graphite has completely finished delithiation and silicon has not started delithiation at all at this point, capacities calculate to $c_{\text{SiO}_x, \text{halfcell}} = 3.3 \text{ mAh}$ and $c_{\text{Gr, halfcell}} = 6.9 \text{ mAh}$ which is 32.4% and 67.6% of the total 10.2 mAh of the composite anode, respectively. Although the assumption above is not entirely true since indeed there is graphite delithiation happening above and silicon delithiation happening below the separation voltage this method still generates highly accurate results for the cells investigated as will be shown in the following section.

For validation, the half-cell voltage curves of pure silicon and pure graphite electrodes are used. These curves are adjusted according to the capacities $c_{\text{SiO}_x, \text{halfcell}}$ and $c_{\text{Gr, halfcell}}$ estimated by DVA. The sum of the individual capacities of SiO_x and graphite at a given voltage level equals the theoretical capacity of the composite anode. Fig. 6(c) and (d) show the comparison of the measured and adjusted voltage curves of SiO_x and graphite, the calculated voltage curve of the composite anode and the actual measurement for the composite anode for lithiation and delithiation separately. For lithiation the calculated and measured voltage curves for the composite anode fit almost perfectly, indicating the validity of estimating the capacities of both active materials using a single distinct peak in the DVA of discharge. For delithiation, both curves show good agreement until silicon starts to delithiate and a slight deviation becomes visible. In this case it is evident that this deviation is caused by the capacity loss observed in the half-cell measurements. This calculation also suggests that the capacity loss was caused by SiO_x since the curves show deviation only in the area where SiO_x almost exclusively contributes to the anode capacity.

With validation on half-cell level the procedure of capacity estimation can also be applied on pouch cell level. Fig. 6 (e) & (f) show pouch cell voltage curves and DVA of discharge for cycle 23. The DVA looks different to that of the half-cells as on pouch cell level the cathode introduces additional peaks. However, the distinct silicon peaks can

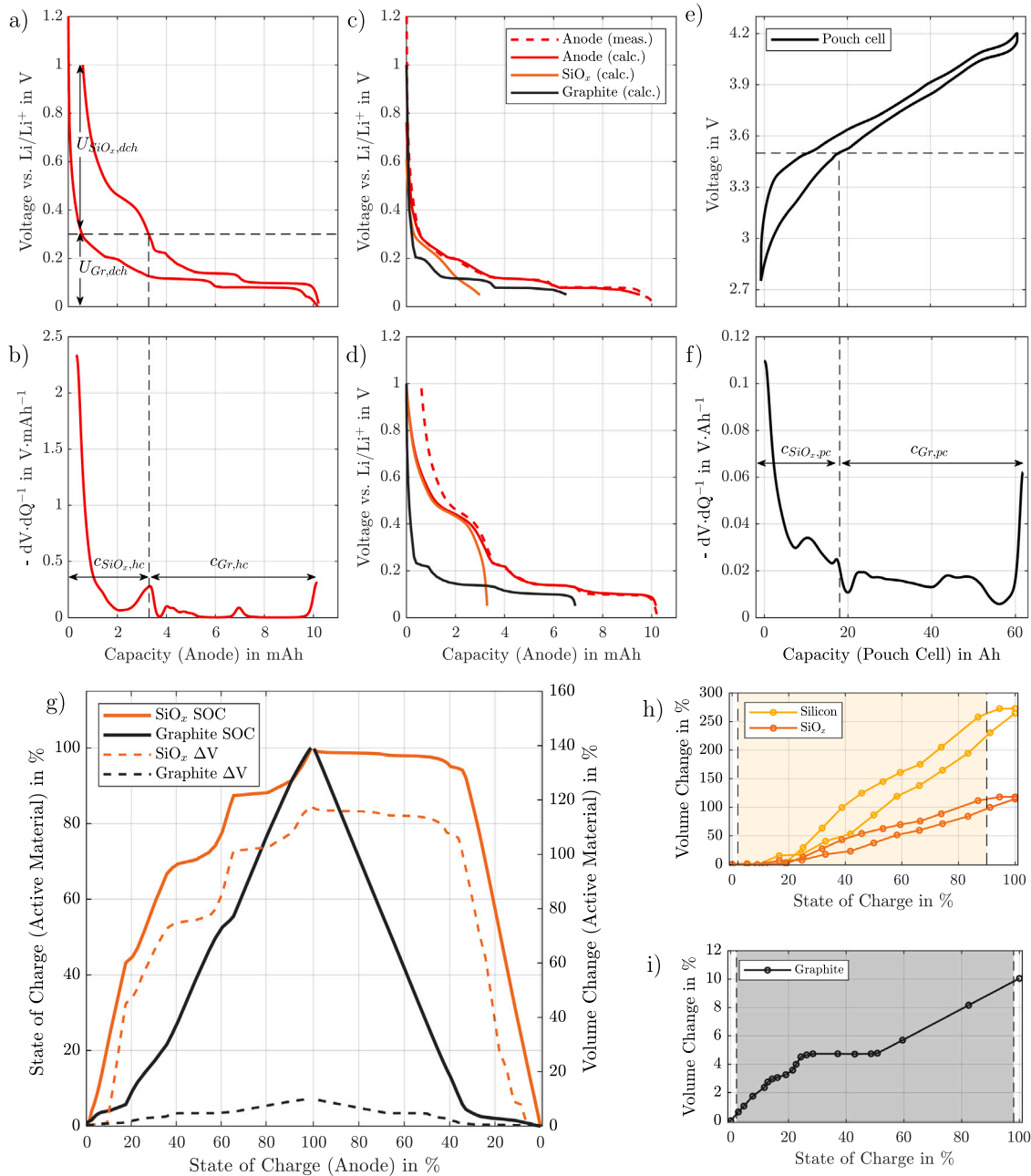


Fig. 6. (a) Measured anode half-cell voltage curve and (b) DVA of discharge. The distinct silicon peak is used to estimate the capacity contribution of each active material. (c) Measured anode voltage curve for delithiation and calculated voltage curves of SiO_x , graphite and the composite anode for lithiation and (d) for delithiation. The theoretical anode curve shows good agreement with the measured values. For delithiation an offset is visible at high voltage because of capacity loss on half-cell level. (e) Pouch cell voltage curve of cycle 23 and (f) DVA of discharge. (g) State of charge and volume change of SiO_x and graphite as a function of the SOC of the composite anode calculated from theoretical voltage curves (c) and (d). Percentage volume change of (h) pure silicon [46] and SiO_x (based on [19,46]) and (i) graphite [47] with respect to the SOC of each active material.

still be spotted easily. The resulting SiO_x capacity on pouch cell level $c_{\text{SiO}_x, \text{pouchcell}} = 17.3 \text{ Ah}$ equals 28.0% of the total 61.7 Ah which is lower than the 32.4% of the total 10.2 mAh on half-cell level. This is most likely caused by the differences in C-rates (C/120 in half-cells vs. C/10 in cell press) as silicon has lower conductivity and diffusion coefficients than graphite [19]. We have observed this general trend in several additional experiments, e.g. 29.4% SiO_x -capacity for a pouch cell cycle with C/20. This implies operating SiO_x -containing cells at varying C-rates has influence on the capacity ratio of SiO_x and graphite, which in turn influences the reversible thickness change.

Particle volume expansion as a function of anode SOC Fig. 6(d) shows how the voltage curves of SiO_x and graphite contribute to the combined voltage curve of the composite anode. In other words, for

any given arbitrary voltage U_1 the capacity of each active material and the combined capacity of the composite anode is known [36]. The capacities $c_{\text{SiO}_x}(U_1)$, $c_{\text{Gr}}(U_1)$ and $c_{\text{Anode}}(U_1)$ are divided by the maximum capacity of the respective component $c_{\text{SiO}_x, \text{max}}$, $c_{\text{Gr}, \text{max}}$ and $c_{\text{Anode}, \text{max}}$ to calculate the SOC of each component as a function of the voltage U_1 according to Eq. (4).

$$\begin{aligned} \text{SOC}_{\text{SiO}_x}(U_1) &= \frac{c_{\text{SiO}_x}(U_1)}{c_{\text{SiO}_x, \text{max}}} \quad \text{and} \quad \text{SOC}_{\text{Graphite}}(U_1) = \frac{c_{\text{Gr}}(U_1)}{c_{\text{Gr}, \text{max}}} \\ \text{SOC}_{\text{Anode}}(U_1) &= \frac{c_{\text{Anode}}(U_1)}{c_{\text{Anode}, \text{max}}} = \frac{c_{\text{SiO}_x}(U_1) + c_{\text{Gr}}(U_1)}{c_{\text{SiO}_x, \text{max}} + c_{\text{Gr}, \text{max}}} \end{aligned} \quad (4)$$

This calculation is done for the whole voltage range $0.05 \text{ V} < U_1 < 0.90 \text{ V}$ for charging and discharging to derive a correlation

between the SOC of each component as a function of voltage. The results allow visualizing the SOC of any component as a function of the SOC of another component. Fig. 6(g) shows the SOC of SiO_x and graphite as a function of the SOC of the anode. As expected, large hysteresis is observed between charge and discharge for SiO_x while graphite shows fairly symmetrical behavior. Previous studies have reported the correlation between voltage, SOC and the percentage volume expansion of pure silicon [36,46] and graphite [36,47]. According to this data the reversible volume change due to lithiation is about 260% for pure silicon and 10% for graphite as shown in Fig. 6(h) and (i). The anodes in this study however contain SiO_x , which experiences significantly less volume expansion than pure silicon. Pan et al. measured a volume change of 118.2% when cycled between 1.50 V and 0.005 V [19] which is close to the theoretical value of 117.0% calculated by Obrovac and Chevrier [3]. To derive the volume expansion of SiO_x as a function of its own SOC it is assumed that the qualitative volume expansion of SiO_x is similar to that of pure silicon. We believe this is a reasonable assumption since silicon is the only electrochemically active component in SiO_x [18,19,48]. The absolute values for pure silicon reported by Beaulieu et al. [46] are scaled to match the value of 118.2% reported for SiO_x by Pan et al. [19] as shown in Fig. 6(g). The data in Fig. 6(h) and (i) is used to calculate the volume expansion of SiO_x and graphite as a function of the SOC of the composite anode as shown in Fig. 6(h). Since none of the active materials was fully lithiated or delithiated only the highlighted regions that correspond to the cycling range of 0.90 V–0.05 V (Fig. S1) of the anode were used. For graphite the volume expansion as a function of the composite anode SOC looks similar to the volume expansion with respect to its own SOC. This is because the SOC of graphite is a fairly linear function of the composite anode SOC. For SiO_x it is the exact opposite, the volume expansion as a function of its own SOC is fairly linear while the SOC is a highly nonlinear function of the composite anode SOC with a large hysteresis. This leads to a volume expansion of SiO_x that looks similar to the SOC as a function of the composite anode SOC.

To investigate the correlation between the volume expansion on the active materials and thickness change of the electrode, the influence of binder, carbon black and SEI are assumed to be neglectable in comparison to the influence of the active materials. Previous studies have confirmed that this is a reasonable assumption [31,40,49]. In case of calendared electrodes with flake-shaped active material particles the graphite particles and graphene layers tend to lay perpendicular to the current collector [38]. This causes the main share of the volume expansion to expand towards the direction of the electrode thickness rather than to expand into the surrounding pores. In a theoretical scenario of constant electrode porosity and free volume expansion a doubling in active material volume would lead to a doubling in electrode thickness (Eq. (5)).

$$\Delta h_{\text{Electrode}} = \Delta V_{\text{AM},\%} h_{\text{Electrode}} \quad (5)$$

Here, $\Delta V_{\text{AM},\%}$ is the percentage volume change of the active material. However, if there are mechanical constraints and pressure on the electrode surface or if particles expand into pores the percentage volume change of the active materials does not equal the percentage thickness change of the electrode [50]. For example in the theoretical scenario of perfectly round shaped particles only the third root of the volume expansion is directed towards the electrode thickness (Eq. (6)).

$$\Delta h_{\text{Electrode}} = \sqrt[3]{\Delta V_{\text{AM},\%}} h_{\text{Electrode}} \quad (6)$$

Previous studies assumed that the absolute pore volume remains constant [37,50] which means that the porosity decreases upon lithiation. Sauerteig [31] and Rieger [40] were able to show that Eq. (7) produced realistic results based on this assumption for electrodes with a single active material.

$$\begin{aligned} \Delta h_{\text{Electrode}} &= \epsilon_{\text{AM}} \Delta V_{\text{AM},\%} h_{\text{Electrode}} = \frac{V_{\text{AM}}}{V_{\text{Electrode}}} \Delta V_{\text{AM},\%} h_{\text{Electrode}} \\ &= \frac{\Delta V_{\text{AM}}}{A_{\text{Electrode}}} \end{aligned} \quad (7)$$

Here, ϵ_{AM} is defined as the quotient of the active material volume V_{AM} and the total electrode volume $V_{\text{Electrode}}$. Vidal et al. [37] have calculated the theoretical porosity change with a similar approach. However they did not differentiate between which of the active materials actually caused the change of porosity and to what degree. In the following section a model fit is performed which aims to distinguish between the influence of graphite and SiO_x on the anode thickness and porosity, respectively.

Model Fit

First, Eq. (7) is expanded to include multiple active materials, in this case SiO_x and graphite. This calculation assumes both active materials expand as if they were the only active material and no interaction takes place.

$$\Delta h_{\text{Anode}} = \Delta h_{\text{SiO}_x} + \Delta h_{\text{Gr}} = \frac{\Delta V_{\text{SiO}_x}}{A_{\text{Anode}}} + \frac{\Delta V_{\text{Gr}}}{A_{\text{Anode}}} \quad (8)$$

Fig. 7(a) shows the theoretical thickness change of the anode calculated by Eq. (8) as a function of the composite anode SOC and the actual measurements. The overall shape of the curve correctly captures the distinct characteristics like the beginning and the end of graphite plateau but Eq. (8) overestimates the absolute values of the anode thickness change. As this approach assumes the total pore volume to be constant the overestimation implies that in this case the total pore volume instead decreased. To generate a better fit proportionality constants are added to Eq. (8) which assume a linear correlation between the decrease in pore volume and active material expansion, $p_{\text{SiO}_x} < 1$ and $p_{\text{Gr}} < 1$. That means p_{SiO_x} and p_{Gr} are the fractions of the volume expansions that lead to thickness change for the respective material, $(1-p_{\text{SiO}_x})$ and $(1-p_{\text{Gr}})$ are therefore the fractions of the volume expansions that decreases the absolute pore volume.

$$\Delta h_{\text{Anode}} = p_{\text{SiO}_x} \frac{\Delta V_{\text{SiO}_x}}{A_{\text{Anode}}} + p_{\text{Gr}} \frac{\Delta V_{\text{Gr}}}{A_{\text{Anode}}} \quad (9)$$

Due to the hysteresis of silicon it is possible to separately approximate the values of p_{SiO_x} and p_{Gr} . Graphite has a fairly symmetric profile with the plateaus at charge and discharge at approximately the same height. The difference of the height at which the graphite plateau occurs in the composite anode is caused by the volume expansion of SiO_x while charging and the lack of volume shrinkage of SiO_x at the beginning of discharging. In this case, the height difference $\Delta h_{\text{plateaus,measured}}$ between the charge-plateau and discharge-plateau as measured in the dilatometer is about 2.22%. This is approximately the thickness change that SiO_x causes between where the charge plateau occurs (50% SOC) and the end of charging (100% SOC), which then does not reverse when discharging back to 50% SOC. Following Eq. (9) and Fig. 7(a) the thickness change between these SOC for SiO_x was calculated to 2.75%. Thus p_{SiO_x} is calculated according to Eq. (10).

$$p_{\text{SiO}_x} = \frac{\Delta h_{\text{plateaus,measured}}}{\frac{\Delta V_{\text{SiO}_x,100\%}}{A_{\text{Anode}}} - \frac{\Delta V_{\text{SiO}_x,50\%}}{A_{\text{Anode}}}} = \frac{2.22}{2.75} = 0.8 \quad (10)$$

This means only 80% of the thickness change calculated under the assumption of constant pore volume was actually measured for SiO_x . From 100% SOC until after the discharge-plateau nearly all of the thickness change is caused by graphite. So, the difference between the thickness of the fully charged anode and the discharge-plateau $\Delta h_{\text{charged-plateau,measured}}$ is about equal to the thickness change induced by graphite only and was measured to 2.16% in the dilatometer. Following Eq. (7) the thickness change between these SOC for graphite was calculated to 2.80%. Again p_{Gr} is calculated according to Eq. (11)

$$p_{\text{Gr}} = \frac{\Delta h_{\text{charged-plateau,measured}}}{\frac{\Delta V_{\text{Gr},100\%}}{A_{\text{Anode}}} - \frac{\Delta V_{\text{Gr},50\%}}{A_{\text{Anode}}}} = \frac{2.16}{2.80} = 0.77 \quad (11)$$

Note that $p_{\text{SiO}_x} \approx p_{\text{Gr}}$ which implies the reduction in pore volume is only dependent on the absolute volume expansion and independent on which of the active materials caused the expansion. Fig. 7(b) shows the

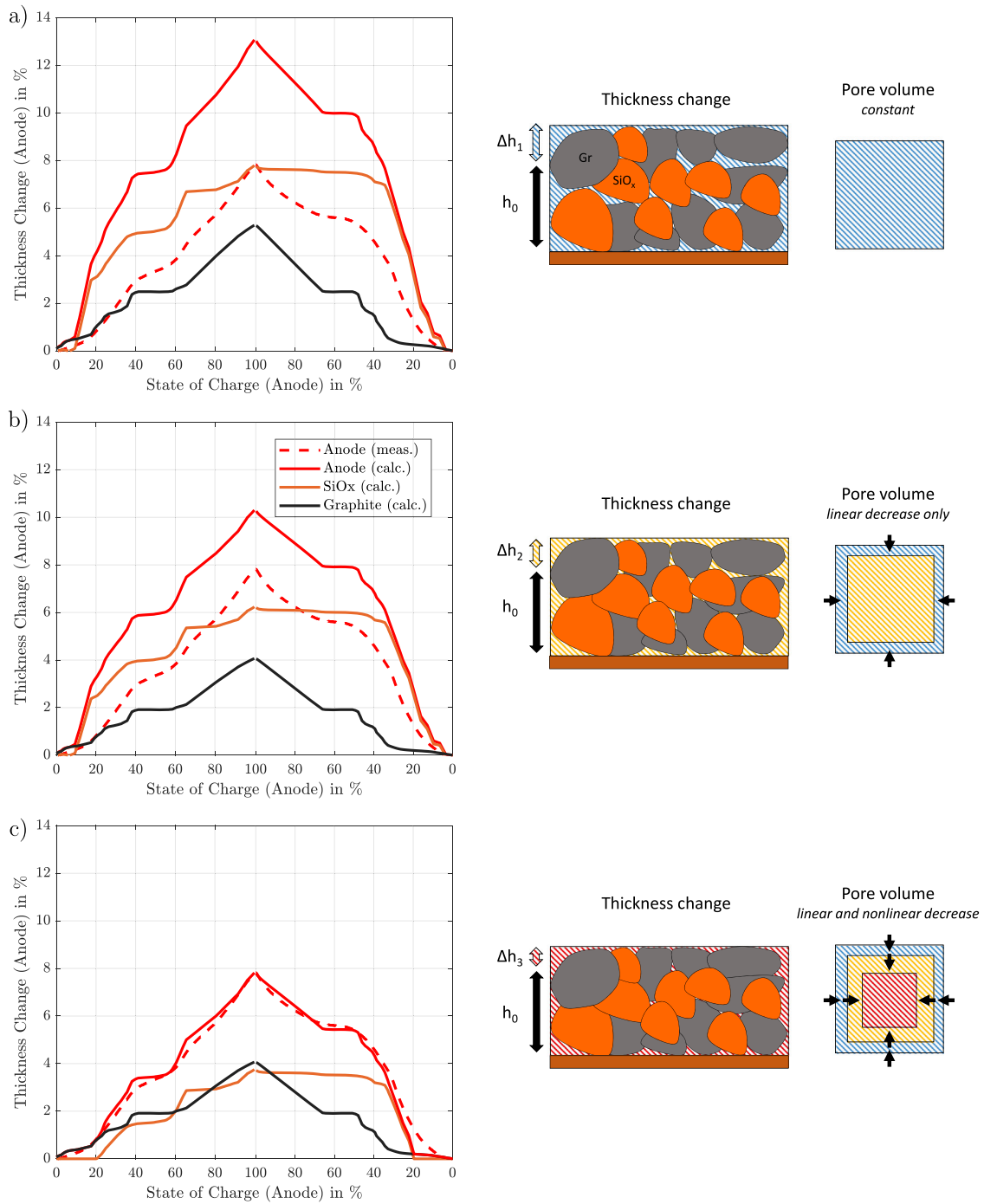


Fig. 7. Theoretical contribution of SiO_x and graphite to the thickness change of the composite anode as a function of anode SOC and comparison with the dilatometer measurements. (a) Contributions calculated according to Eq. (8) which is based on the assumption of constant absolute pore volume. (b) Contributions calculated according to Eq. (9) that includes proportionality constants that account for a linear decrease in absolute pore volume. (c) Contributions calculated according to Eqs. (9) and (12) that accounts for no visible influence of SiO_x below 20% SOC.

theoretical thickness change of the anode as calculated in Fig. 7(a) with the additional proportionality constants for pore volume change. The calculated absolute values fit the measured values much better with a constant offset between calculated and measured values except for low SOC. For charging, the model suggests a very strong contribution to the anode thickness change from the volume expansion of SiO_x below ≈ 20% SOC. However, the measured anode curve in Fig. 7(b) shows no indication of that at all. Similar observations were made in previous studies with Si-Gr composite anodes [36,37]. As Eq. (9) assumes a linear correlation between volume expansion and pore volume decrease. This suggests an additional nonlinearity influences the

thickness change. Interestingly, for low SOC the measured anode curve follows the shape of the calculated graphite curve very closely with no influence of SiO_x at all. Above ≈ 20% SOC the measured anode curve starts to deviate from the graphite curve, indicating that from this point onwards SiO_x starts to influence the thickness change. It is suspected that the elasticity and overall morphology of the surrounding materials cannot follow the continuous large volume changes of the SiO_x particles. A similar phenomenon has been observed by Müller et al. [51] who found significant gaps between SiO_x-particles and the surrounding carbon black-binder-domain inside SiO_x-Gr composite anodes. The gaps were created after the first lithiation and remained as void space

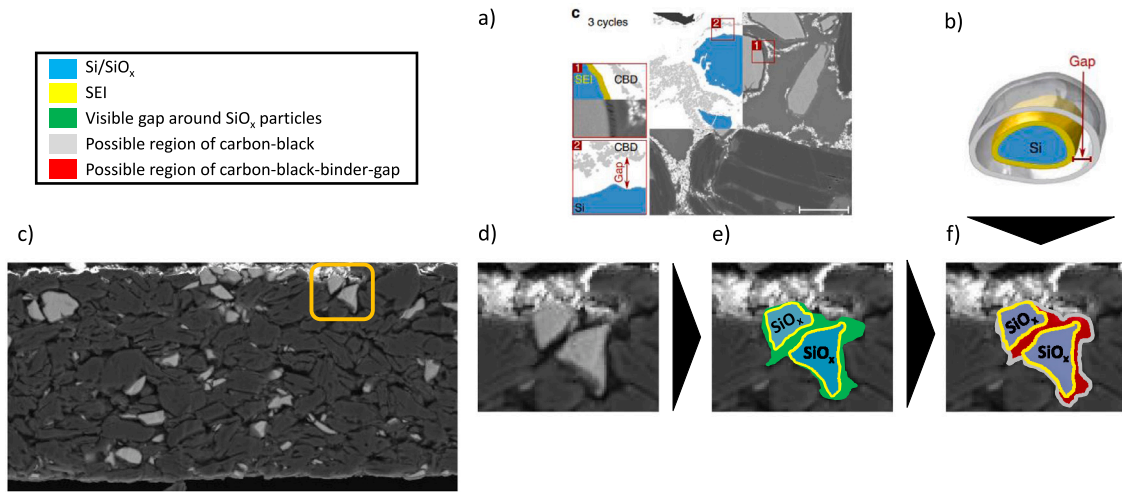


Fig. 8. Carbon black-binder-gap. (a) SEM by Müller et al. that proves the existence of the carbon black-binder-gap around SiO_x particles in SiO_x-Gr anodes [51], adapted with permission from Springer Nature, (b) graphical illustration of the carbon black-binder-gap from Müller et al. [51], adapted with permission from Springer Nature, (c) SEM of the cross section of the SiO_x-Gr anode investigated in this work, (d) zoomed view of the domain around the SiO_x particles (orange square), (e) zoomed view with marked visible gap around the SiO_x particles, note that the carbon-black-binder-domain is not visible here, (f) zoomed view with illustration of the possible region of the carbon black-binder-gap around the SiO_x particles inside the anode investigated in this study based on the findings of Müller et al. [51]. The carbon black-binder-gap is a possible explanation for the delayed influence and the nonlinear reduction in pore volume modeled with Eq. (12). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

between the SiO_x particles and the carbon black-binder-domain at low SOC. To integrate this effect into the model the thickness change induced by SiO_x for < 20% SOC is set to zero (Eq. (12)).

$$\Delta h_{\text{SiO}_x} = 0 \text{ for } \text{SOC} < 20\% \quad (12)$$

With this addition, the calculated curve shows very good agreement with the measured curve. The absolute height of the plateaus and the total thickness change of the composite anode are calculated with almost no error. At the end of discharge for very low SOC there is a slight deviation between calculated and measured values. An influence of void space between active material and the surrounding binder skeleton on the thickness change of electrodes was already theorized but not proven [38]. Fig. 8 shows a possible visualization of this phenomenon based on the findings of Müller et al. (Fig. 8(a) and (b), [51]) and a SEM scan of the anode investigated in this work (Fig. 8(c)). Void spaces around the SiO_x particles are clearly visible (Fig. 8(d) and (e)). However, the carbon black-binder-domain is not visible in a regular SEM scan due to weak signals from the light elements contained in the carbon black-binder-domain and weak image contrast to the graphite particles [51]. Fig. 8(f) shows an illustration of the possible region of the carbon black-binder-gap based on (a) and (b). We can give an estimate for the void volume between the SiO_x particles and the carbon black-binder-domain of $V_{\text{Void}} = 0.47 \cdot V_{\text{SiO}_x}$ which equals the volume change of SiO_x in Fig. 6(g) until it starts to contribute to the thickness change at 20% SOC in Fig. 7(c). When modeled as a single perfect sphere the length of the gap $g_{\%}$ as a percentage of the SiO_x particle radius calculates as shown in Eq. (13).

$$g_{\%} = \sqrt[3]{\frac{V_{\text{Void}} + V_{\text{SiO}_x}}{V_{\text{SiO}_x}}} = 13.7\% \quad (13)$$

On average the radius of the SiO_x particles increased by 13.7% before starting to push against the surrounding morphology and affecting the anode thickness. Müller et al. measured the gap between SiO_x particles and the nearest carbon black-binder-domain with a ray-tracing approach based on segmented tomography data [51]. They found that the gap increased with increasing cycle number and ranged between 23% of the radius for a particle with 2 μm radius and 13% of the radius for a particle with 6 μm radius after the first cycle.

Overall, the fact that Eq. (9) with Eq. (12) calculates the thickness change of the composite anode with little error clearly suggests that the

large volume expansion of SiO_x below 20% SOC is indeed swallowed by pore volume decrease and we believe the SiO_x-carbon black-binder-gap is a reasonable explanation for this phenomenon. Without this effect the thickness change of a single anode coating would be 10.26% (Fig. 7(b)) instead of 7.85% (Fig. 7(c)). This would translate to a total anode thickness change of 4.83% with respect to the pouch cell thickness at 0% SOC in Fig. 4 and a pouch cell thickness change of 4.0% rather than 2.87%. As this additional thickness change would result in larger compression forces induced by the stiffness of the module housing an increased capacity fade would be expected without this effect [26,31]. For SOC > 20% the pore volume continues to decrease according to Eq. (14).

$$\Delta V_{\text{Pores, >20\%SOC}} = -((1 - p_{\text{SiO}_x})\Delta V_{\text{SiO}_x} + (1 - p_{\text{Gr}})\Delta V_{\text{Gr}}) \quad (14)$$

This seems to be largely independent on the type of active material as $p_{\text{SiO}_x} \approx p_{\text{Gr}} \approx 0.8$. The deviation between calculated and measured values at the end of discharge might indicate that the change in pore volume is not symmetrical between charging and discharging but also shows some form of ‘porosity-hysteresis’. Looking at the total anode thickness change of 7.85%, about 4.11% or 52.40% was caused by graphite and about 3.74% or 47.60% was caused by SiO_x.

4. Conclusion

A large-format Li-ion pouch cell with NMC811 cathodes and SiO_x-Gr composite anodes was cycled under constant pressure and constant temperature. The measured reversible cell thickness change showed good repeatability after some initial irreversible thickness change. At the beginning of discharge, the pouch cell thickness first decreased and suddenly increased again between 90% and 80% SOC. Electrochemical dilatometry revealed that this increase in cell thickness was caused by a steep increase in cathode thickness at the beginning of discharge. It is important to note that unlike other cathode materials Ni-rich NMC causes significant thickness change on pouch cell level. In this case, the decrease of NMC811 volume counteracted the thickness change of the SiO_x-Gr composite anode by 21.8%. This is interesting for automotive battery and cell design decisions as this makes it easier to operate the cells within the optimal stress level range for maximum cycle life. Overall, a highly nonlinear expansion behavior for the cathode was observed which was linked to the change of the lattice parameters of

the crystal structure. The comparison between the volume change of the NMC811 unit cell and the thickness change of the cathode suggests that a large portion of the volume change leads to porosity change rather than thickness change. Pouch cell thickness showed large hysteresis which was caused by the voltage and volume hysteresis of SiO_x inside the anode. For closer inspection, the capacities of both anode active materials were estimated using DVA. The capacities were then used to calculate the voltage curves of SiO_x and graphite as a function of the anode SOC. The sum of both active material voltage curves showed great agreement with the actual half-cell measurements of the composite anode and allowed to calculate the SOC and the volume expansion of each active material as a function of the anode SOC. The correlation between the volume expansion and the thickness change of the anode was investigated starting with the theory of constant pore volume and decreasing porosity. The resulting values overestimated the thickness change which indicated that the pore volume did not stay constant but rather decreased. Additional proportionality constants were introduced to account for this behavior and enabled a better fit. In this case, the thickness change caused by each active material was about 80% of what was calculated by the theory of constant pore volume. Additionally, these calculations suggested a large influence on the cell thickness by SiO_x below 20% SOC. However, no such strong influence was actually measured. This indicated that the initial large volume expansion of SiO_x is swallowed by pore volume, which is believed to be located between the SiO_x particles and the carbon black-binder-domain. After integrating this phenomenon into the model a very good fit to the measured data was achieved. Here, the impact of the microscopic SiO_x -carbon black-binder-gaps on the thickness change of large-format Li-ion cells was quantified for the first time. Similar to the collapse of the lattice parameter c of the cathode this effect further makes it easier to ensure optimal pressure conditions through decreased reversible thickness change on pouch cell level. Overall, about 52.40% of the total 7.85% thickness change of the anode was caused by graphite and 47.60% was caused by SiO_x .

CRediT authorship contribution statement

Hendrik Pegel: Conceptualization, Methodology, Software, Validation, Investigation, Writing – original draft, Visualization. **Otto von Kessel:** Conceptualization, Methodology, Validation, Investigation, Writing – original draft, Resources, Visualization. **Philipp Heugel:** Conceptualization, Methodology, Validation, Investigation, Writing – original draft, Visualization. **Tobias Deich:** Methodology, Writing – review & editing. **Jens Tübke:** Supervision, Writing – review & editing. **Kai Peter Birke:** Supervision, Writing – review & editing. **Dirk Uwe Sauer:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thank Margaryta Paramonova and Matthias Hahn from EL-Cell GmbH for performing the dilatometer measurements and providing technical expertise as well as Lisa Willenberg for valuable scientific discussion. Wolfgang Märkle and Matthias Storch are thanked for opening of the cells.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jpowsour.2022.231443>.

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